Near-infrared spectroscopy for collagen prescreening in archaeological bone: a review and implications for a Spanish measurement campaign

Nov 2, 2026 · @Alejandro

Abstract

Near-infrared (NIR) spectroscopy can now predict the collagen yield of archaeological bone non-destructively with an error of about 1.6 percentage points, close to the reproducibility of collagen extraction itself. This review synthesises six NIR studies published between 2013 and 2026, from qualitative hyperspectral imaging (Linderholm et al., 2013; Vincke et al., 2014) to quantitative single-point models trained on 140 radiocarbon-dated bones (Ryder et al., 2026). It compares them with four Raman and ATR-FTIR studies of archaeological bone, and adds evidence from an open, reproducible re-analysis of the Ryder et al. dataset.

Four conclusions emerge. First, the collagen signal is carried by protein combination bands at 2030-2060, 2150-2195 and 2270-2300 nm, and a one-component model on 2030-2060 nm is as accurate as any of 3004 alternatives tested under nested, leave-one-country-out validation. Second, a three-wavelength index centred on 2045 nm matches that model within one standard error and is more tolerant of wavelength shifts. Third, the 1900-1960 nm region assigned to hydroxyapatite in one study behaves as water in the larger dataset, so no mineral/collagen ratio could be validated. Fourth, the main limits are not spectral: they are the precision of the reference yields, intra-bone heterogeneity, consolidants, and site offsets of 1-2 percentage points. Of these, polyvinyl acetate can be detected reliably by reading the second-derivative dip at 2137 nm against the 2045 nm collagen band.

We close with a measurement and validation protocol for a new campaign on Spanish assemblages, combining NIR with Raman and ATR-FTIR at the same spots, designed to test these limits prospectively.

Keywords: near-infrared spectroscopy; bone collagen; radiocarbon prescreening; chemometrics; diagenesis; spectral indices

1. Introduction

Bone collagen is the material behind most radiocarbon chronologies of the Late Pleistocene and Holocene, behind stable-isotope reconstructions of diet and mobility, and behind palaeoproteomic identification such as ZooMS (Sponheimer et al., 2019; Ryder et al., 2026). After burial it degrades at rates set by temperature, hydrology, soil chemistry and microbial attack, so preservation varies widely between sites, between bones of one site, and within a single bone (Lugli et al., 2021; Malegori et al., 2023).

A bone is generally considered suitable for radiocarbon dating and isotopic analysis when it yields at least 1 % collagen by weight (van Klinken, 1999, as cited by Sponheimer et al., 2019; Lugli et al., 2021; Ryder et al., 2026). Palaeolithic bones often need around 500 mg of starting material to reach that amount (Malegori et al., 2023). At sites with poor preservation, laboratories may destroy many specimens to find a few that are datable. Sponheimer et al. (2019) note that only three of 50 bones from the Neanderthal site of Zafarraya retained more than 4 % collagen.

The cost is not only financial. Faunal and human collections are finite, and the ethical case against unnecessary destructive sampling has been made forcefully by curators, descendant communities and researchers (Lugli et al., 2021; Ryder et al., 2026). This has driven the search for prescreening methods that are fast, cheap, and ideally non-destructive.

NIR spectroscopy is a strong candidate. It needs no sample preparation, measures in seconds with portable instruments, and probes millimetres rather than microns below the surface (Sponheimer et al., 2019; Ryder et al., 2026). This review has three aims:

1. To synthesise what is known about the NIR signal of collagen in archaeological bone, and how well it predicts collagen yield.
2. To identify the confounders that limit transfer to new sites, using both the literature and a reproducible re-analysis of the largest public dataset.
3. To derive a measurement and validation protocol for a new campaign on Spanish assemblages.

2. Established prescreening methods

NIR is not competing with nothing: three families of methods are already in routine use, and each sets a benchmark for what a new screen must beat.

Method	What it measures	Sample needed	Reported performance	Main limits
Whole-bone %N and C:N	Nitrogen content as a proxy for protein; C:N for quality	About 5 mg, destroyed	Above 0.7-0.76 % N predicts more than 1 % collagen 70-73 % of the time; 72 % correct on 50 Zafarraya bones, 63 % on the 19-bone subset of Ryder et al.	Destructive; inflated by nitrogen-rich contaminants such as decomposition products (Lebon et al., 2016)
ATR-FTIR on bone powder (Lebon et al., 2016)	Amide I / phosphate band-area ratio	Under 1 mg of a 50 mg ground aliquot	Standard error 0.22 % N, about 1.2 % collagen; PLS no better; detects collagen from about 3 % and quantifies from about 4 %	Weak at the 1 % threshold; water and humic bands overlap amide I; replicate spread up to 24 %
ATR-FTIR on extracted collagen (Martínez Cortizas and López-Costas, 2020)	Amide, CH2 and carbohydrate bands of the extract	Extracted collagen	Band structure explains 92 % of collagen quality (C, N, C:N); yield only moderately	Quality control after extraction, not prescreening
FT-Raman, benchtop, 1064 nm (France et al., 2014)	Phosphate (960 cm-1) to amide I (1636 cm-1) peak-height ratio	None on a fresh cross-section	Ratio up to 19.4 marks well-preserved collagen (C:N 2.8-3.6); PLS-DA 95 % correct on cross-sections	Outer surfaces mislead (16-68 % correct); the ratio varies by about 38 % within a bone

Method	What it measures	Sample needed	Reported performance	Main limits
Handheld Raman, 1064 nm (Madden et al., 2018; Pestle et al., 2015)	Phosphate to CH2 (1450 cm-1) or amide I ratio	None	86 % correct on cross-sections with 960/1450; 1450/960 above 0.1 indicates more than 1 % collagen (Pestle et al., as cited)	The 1450 cm-1 band and C-H region are shared by consolidants; amide I is lost in noise
NIR spectroscopy	Overtones and combinations of C-H, N-H and O-H	None	RMSE 1.2-2.2 % within a collection, 1.6-2.8 % at a new site (Section 6)	Needs calibration; sensitive to water, consolidants and instrument

Sources: Lebon et al. (2016); France et al. (2014); Madden et al. (2018); Martínez Cortizas and López-Costas (2020); Sponheimer et al. (2019); Malegori et al. (2023); Ryder et al. (2026). The %N thresholds are those of Brock et al. (2010, 2012) and Jacob et al. (2018), as cited there; Malegori et al. use 0.70 % N for Palaeolithic bone.

The methods are calibrated against different references. Lebon et al. use whole-bone %N, converted to collagen by assuming that modern bone holds about 4 % N for 22 % collagen. France et al. and Madden et al. use the C:N ratio of the extract to classify quality. The NIR studies use the yield of ultrafiltered collagen. Results are therefore not directly comparable, and a new campaign should record all three references (Section 10.4).

The mid-infrared methods share a useful property that NIR lacks: they see the mineral directly through the strong phosphate band, so a collagen/mineral ratio is well defined. Section 3 shows why this does not carry over to NIR.

3. Physical basis: what NIR sees in bone

Bone is a composite of carbonated apatite, Ca5(PO4,CO3)3(OH), a collagen-dominated organic matrix, and water (Vincke et al., 2014). Collagen makes up about 85-90 % of the organic fraction (Ryder et al., 2026). NIR absorption arises from overtones and combinations of vibrations involving hydrogen: C-H, N-H and O-H. Bone is strongly

scattering, so reflectance R is converted to pseudo-absorbance, log10(1/R), rather than true absorbance (Ryder et al., 2026). Light penetrates on the order of millimetres, against microns for ATR-FTIR and Raman (Sponheimer et al., 2019).

The consequence is an asymmetry. Collagen is rich in N-H and C-H and has several specific bands. Phosphate, the bulk of the mineral, has its fundamental vibrations near 1000 cm-1 (about 10 um) and no usable band below 2500 nm. The mineral appears in NIR only through its scattering and through O-H, which it shares with water.

Band (nm)	Main assignment	Reported by	Status for prescreening
1690-1750	C-H stretch, first overtone	Sponheimer 2019; Ryder 2026	Not protein-specific; weaker predictor
1410 / 1430	O-H (water or apatite) / N-H first overtone	Vincke 2014	Dominated by water in the re-analysis (r = 0.89 with the 1930 nm band)
1910-1960	O-H of water (bend + stretch combination)	Sponheimer 2019 (1940, excluded); Vincke 2014 (1910)	Water; see the contested row below
1959	O-H of hydroxyapatite	Lugli 2021	**Contested:** follows the water band at r = 0.94 in the re-analysis
1950-1970	Possibly amine compounds	Linderholm 2013	Tentative
2030-2060 (2040-2045)	N-H stretch + amide II combination; C=O	Sponheimer 2019; Vincke 2014 (2030); Malegori 2023 (2040); Ryder 2026	**Most protein-specific; basis of the preferred models**
2150-2195	N-H bend overtone + C=O stretch (amide I/II combinations)	Sponheimer 2019 (2175); Vincke 2014 (2150); Lugli 2021 (2195); Malegori 2023 (2170)	Specific, but near the PVA peak at 2135 nm
2270-2300	C-H stretch + deformation combinations	Sponheimer 2019; Lugli 2021 (2293); Malegori 2023 (2270)	Overlaps the PVA peak at 2296 nm

Band (nm)	Main assignment	Reported by	Status for prescreening
2135, 2250, 2296	Polyvinyl acetate (consolidant)	Ryder 2026	Exclusion zones

Two cautions apply to every row. NIR bands are broad and overlapping, so each assignment names a dominant contribution rather than a single vibration (Ryder et al., 2026). And the 1900-1960 nm region carries three different assignments in this literature. Section 8 tests them directly.

4. Instrumentation and acquisition

Two instrument families are in use, and they answer different questions: single-point spectrometers ask *which bone*, hyperspectral cameras ask *where in the bone*.

Study	Instrument	Range and resolution	Sample presentation	Reference
Linderholm et al. 2013	sisuCHEMA push-broom camera	1000-2498 nm, 6-7 nm steps	Bones, teeth and sieved sediment on a tray or belt	Not stated in the text we read
Vincke et al. 2014	XEVA camera with MCT detector on a conveyor belt	1100-2400 nm, 6.3 nm steps, 209 channels	Small fragments, both faces, air-dried to equal humidity	Teflon plate
Sponheimer et al. 2019	ASD LabSpec 4, fibre-optic probe	350-2500 nm	Ground bone in glass vials; whole bone at exposed cross-sections	Not stated
Lugli et al. 2021	Specim SWIR3 camera	1000-2500 nm, 5.6 nm; pixel 500 um	Whole bones on a moving stage	99 % Spectralon
Malegori et al. 2023	Specim SWIR3 camera	1000-2500 nm (1060-2500 used), 5.6 nm	Powders and fragments; concretions masked	99 % Spectralon

Study	Instrument	Range and resolution	Sample presentation	Reference
Ryder et al. 2026	ASD LabSpec 4 Standard Resolution, bifurcated probe, 2 mm spot	350-2500 nm; 3 nm (VNIR), 10 nm (SWIR); detector joins at 1000 and 1800 nm	Flat periosteal surface of whole bone, probe flush; 50 scans averaged	Not stated in the main text

Several practical lessons recur. Single-point acquisition takes about 5 s, which allows more than 500 bones a day (Ryder et al., 2026). Repeat scans at one spot vary by only 0.03 % collagen (coefficient of variation 0.85 %), so instrument noise is negligible next to other errors. Hyperspectral imaging takes minutes per bone but reveals heterogeneity that a single spot cannot (Lugli et al., 2021; Malegori et al., 2023).

Surface condition matters in both families. Adhering sediment and concretions have their own NIR bands and must be masked or cleaned (Linderholm et al., 2013; Malegori et al., 2023). Spectra should be taken as close as possible to where collagen was or will be extracted (Ryder et al., 2026).

Models do not transfer freely between instruments. Ryder et al. caution that device-to-device calibration is typically required. The re-analysis in Section 8 quantifies this for a change of spectral resolution.

5. Chemometric approaches

The field has moved from exploration to quantification: from PCA and class membership (2013-2014) to regression on collagen yield (2019-2026), with a parallel thread of simple band indices.

Preprocessing. Savitzky-Golay second derivatives dominate: third-order polynomial with a 31-point window on the LabSpec 4 (Sponheimer et al., 2019; Ryder et al., 2026), and a 15-point window after standard normal variate (SNV) on hyperspectral data (Malegori et al., 2023). Derivatives remove additive baselines and sharpen overlapping bands. SNV corrects multiplicative scatter (Linderholm et al., 2013; Malegori et al., 2023). Vincke et al. (2014) used first derivatives.

Variable selection. Restricting the range is the most consequential choice. Sponheimer et al. combined 1695-1750 and 2000-2300 nm and excluded water bands. Ryder et al. compared ten ranges and preferred 2030-2060 nm with one latent variable: it was near

the best and avoids the PVA peaks.

Models. Four families appear:

- **PLS regression** is the workhorse (Sponheimer et al., 2019; Malegori et al., 2023; Ryder et al., 2026).
- **Random Forest** gave lower calibration errors than PLS in Ryder et al. but did worse at Zafarraya when it used the full range, which the authors attribute to overfitting and PVA-affected variables.
- **PLS-DA** answers the binary question "should I sample this bone?" directly (Vincke et al., 2014; Sponheimer et al., 2019).
- **Band indices** use two or three reflectances. Lugli et al. (2021) proposed a normalised difference image (NDI) for mapping:

$$\mathrm{NDI} = \frac{R_{1959} - R_{2195}}{R_{1959}}$$

The re-analysis proposes a three-band continuum index centred on the 2045 nm band (Section 8):

$$\mathrm{CI}_{2045} = \frac{R_{2020} + R_{2075} - 2R_{2045}}{R_{2020} + R_{2075} + 2R_{2045}}$$

A two-band ratio cancels brightness but not a sloping baseline. Adding a third band, as a continuum on both sides, removes the slope as well. This is the discrete analogue of the second derivative used by the PLS models.

Validation practice has improved but remains uneven. Early studies split samples by sorting on yield and alternating them between calibration and validation (Sponheimer et al., 2019). Two studies validated against independent material: Malegori et al. (2023) on six bones extracted after prediction, and Ryder et al. (2026) at a separate site. Malegori et al. also report a permutation test, in which RMSECV rose from 2.2 % to about 6 % with the labels shuffled. None of the published studies validated across sites in cross-validation, and none corrected for the optimism of choosing among many models. Both matter, as Section 8 shows.

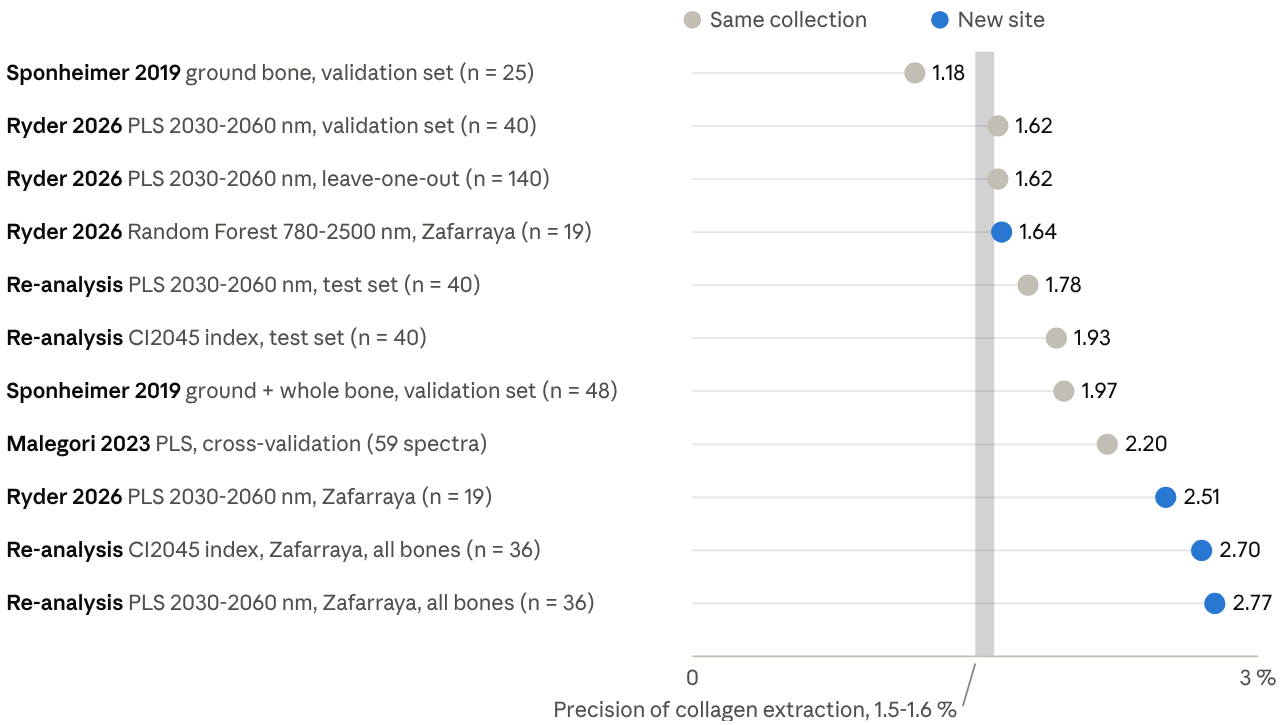
6. Reported performance

Quantitative NIR models predict collagen yield with errors of 1.2-2.2 % within the collection they were trained on, and 1.6-2.8 % at a new site. The studies differ in nearly every design choice, so the numbers are comparable in scale, not in rank.

Study	Bones	Age	Sample form	Collagen reference	Model	Validation
Linderholm et al. 2013	Elk bones and teeth, one site (Bastuloken, Sweden)	about 6000-2200 BP	Whole bone, sieved sediment	None	PCA	Qualitative
Vincke et al. 2014	16 calibration + 4 validation, one site (Trou Al'Wesse, Belgium)	Mesolithic to Neolithic	Fragments, both faces	AMS success and stratigraphy (classes)	PLS-DA, 1st derivative	200 held-out spectra, 100 % correct at spectrum level
Sponheimer et al. 2019	50 ground; 97 ground + whole	about 500 to 45,000 years	Ground and whole bone	Modified Longin and ultrafiltration (two labs)	PLS, PLS-DA	Alternating split sorted by yield
Lugli et al. 2021	6 reference + 3 test, Italy	7th c. AD to 45,000 BP	Whole bone, imaged	HCl demineralisation, no ultrafiltration	NDI index	Box plots; targeted re-extraction
Malegori et al. 2023	44 bones (59 calibration spectra)	Modern to over 50,000 years	Powders and fragments	Ultrafiltration, >30 kDa fraction	PLS, SNV + 2nd derivative	5-group CV, permutation test, 6 external bones
Ryder et al. 2026	140 (from 319) + 36 Zafarraya	about 500 to 45,000 years	Whole bone, periosteal surface	Acid-base-acid + ultrafiltration (MPI, ORAU)	PLS, Random Forest	100/40 split, leave-one-out, external site

Reported errors cluster near extraction precision; most new-site errors exceed 2.5 %

RMSE of predicted collagen yield, percentage points, by study and evaluation design



Values as reported by each study and by the re-analysis notebooks (Section 8). Designs, sample forms and reference protocols differ; see the table above.

The re-analysis reproduces the published 100/40 split, but its test error for the published PLS (1.78 %) is higher than the 1.62 % reported. Differences in software and in flooring predictions at zero may explain part of the gap; we could not reconcile it fully. The Random Forest result at Zafarraya (1.64 %) is low in error but classified only 11 of 19 bones correctly at the 3 % threshold, a reminder that RMSE and screening success are different targets (Ryder et al., 2026).

7. Sources of error and confounders

The dominant errors in NIR collagen prediction are not spectral noise but the reference values, the sample and the site. Six sources recur across the literature.

7.1 Precision of the reference yield. Collagen extraction itself varies by about 1.5 % between laboratories (Pestle et al., 2014, as cited by Ryder et al., 2026) and 1.6 % within a laboratory (Sponheimer et al., 2019). No model can be expected to predict yield more precisely than it can be measured. Published errors already sit at this level (Figure in Section 6).

7.2 Intra-bone heterogeneity. Collagen content varies by a factor of two to five within one bone: 12.5 % against 25 % in one femur, and 5.3 % against 26.4 % in one coxal bone (Lugli et al., 2021). Areas in contact with the sediment tend to be more degraded (Malegori et al., 2023). Because NIR spectra are usually acquired after extraction, the scanned spot may not represent the extracted portion. Ryder et al. (2026) attribute part of their Zafarraya error to this mismatch. Other methods see the same problem: Raman mineral-to-collagen ratios differ by about 38 % between spots on one cross-section and by 32 % between outer surface and cross-section (France et al., 2014), and triplicates of homogenised powder differ by up to 43 % in %N and 24 % in the FTIR amide I / phosphate ratio (Lebon et al., 2016).

7.3 Definition of yield. Protocols differ in demineralisation, base treatment and filtration. Ultrafiltration keeps only the fraction above 30 kDa; in one validation bone this was 7 % of the bone against 9.2 % in total (Malegori et al., 2023). Models trained on ultrafiltered yields are likely to under-predict yields from other protocols (Ryder et al., 2026).

7.4 Water. Liquid water absorbs strongly near 1450 and 1940 nm, and its content varies with burial, storage and humidity. Vincke et al. (2014) found that the first principal component of their data loaded on O-H bands at 1410 and 1910 nm rather than on collagen. Sponheimer et al. (2019) excluded the water band from their models. Water also affects the mid-infrared: its bending band near 1635 cm-1 lies inside amide I, so ATR-FTIR powders must be dried consistently (Lebon et al., 2016). Section 8 shows that a band assigned to the mineral (Lugli et al., 2021) behaves as water across many sites.

7.5 Consolidants and their removal. Polyvinyl acetate absorbs at 2135, 2250 and 2296 nm, close to two collagen bands. Treated bones appear as outliers in Hotelling's T2, and two untreated-looking bones in Ryder et al. carried strong PVA bands. Solvent washing also degrades collagen: treated Zafarraya bones yielded 1.05 % against 1.89 % for untreated ones (Ryder et al., 2026). Models restricted to 2030-2060 nm avoid the known PVA peaks but not unknown consolidants. Section 8.8 gives a detection rule and a correction for PVA. Consolidants also bias Raman screening: PVA, Paraloid B-72, cellulose nitrate and soluble nylon all have bands at 1450 cm-1 and in the C-H region, so only the amide I band at 1636 cm-1 is protein-specific (Madden et al., 2018).

7.6 Site and instrument. Every study that tested a new site reported a higher error than in calibration (Section 6). Burial environment is one likely driver. Among 50 skeletons from eight sites in NW Spain, collagen was better preserved in alkaline, well-drained or confined contexts (a limestone cave, coastal palaeodunes) than in acidic soils, while variation within each site exceeded variation between periods (Martínez Cortizas and

López-Costas, 2020). Device-to-device differences in wavelength calibration and resolution also require recalibration (Ryder et al., 2026). Section 8 measures both effects on the public dataset.

8. Evidence from a reproducible re-analysis

We re-analysed the public spectra of Ryder et al. (2026) to test which of the claims above hold under stricter validation. The data are 140 reference bones and 36 Zafarraya bones, measured on the LabSpec 4 over 350-2500 nm. The work is a sequence of eight Python notebooks (`model_exploration/` in the project repository) with numbered decisions and cached results.

8.1 Validation design. We kept the published 100/40 split and scored every candidate on the 100 calibration bones by J, the mean of two cross-validated RMSEs: repeated stratified 5-fold, and leave-one-country-out. The country term stands in for a new site, which is the use case of prescreening. The model search was itself nested inside cross-validation to measure the optimism of picking a winner. Test and Zafarraya data were opened only after the choices were frozen.

8.2 No model beats the published one. A search over 3004 candidates covered 11 preprocessing recipes, 8 wavelength regions, PLS variants, Gaussian processes, support vectors and forests. Nested inside cross-validation, the search procedure scored J = 1.88-2.14 %, against 1.64 % for the published one-component PLS on 2030-2060 nm.

Model	Random CV (%)	Country CV (%)	Test, n = 40 (%)	Zafarraya, n = 36 (%)	Largest shift under synthetic PVA (%)
PLS 2030-2060 nm, 1 LV (published, recommended)	1.54	1.76	1.78	2.77	0.00
Random Forest 2030-2060 nm (published)	1.51	1.97	1.62	3.15	0.00
Random Forest 780-2500 nm (published)	1.34	2.01	1.80	2.62	4.26
Search winner, PVA-safe regions	1.39	1.71	1.59	2.23	0.83

The search winner's CV values are optimistic by construction. None of the test-set differences from the published PLS is statistically resolved.

8.3 Site offsets are real. At equal band depth, Spanish bones (46 of the 100 calibration bones) yield 1.44 % less collagen than the others (95 % bootstrap interval for the three-band index: -2.03 to -0.90 %). No preprocessing or band combination removed it. At Zafarraya the published PLS over-predicted by +1.5 % on average, and CI2045 by +1.3 %. Subtracting the mean error of five locally extracted bones reduced the PLS error from 2.77 to a median of 2.15 %.

8.4 Triage rather than quantification. Thresholding the regression worked as well as a separate classifier. Predictions below 1.75 % are labelled Skip, from 2.95 % Sample, and Check in between. On the test set, Skip held 11 bones of which 2 were datable; Sample held 20, all datable.

8.5 A three-wavelength index. CI2045 (Section 5) correlates with yield at r = 0.94. Its nested J is 1.68 %, tied with PLS within one standard error (0.15 %). On the test set it is measurably less accurate (RMSEP 1.93 against 1.78 %; paired difference 0.05-0.25 %). At Zafarraya it ties (2.70 against 2.77 %). Its strength is robustness: a 2 nm wavelength shift moves its predictions by 0.02 % on average, against 0.12 % for PLS. Narrower or searched indices scored better in calibration but were 15-30 times more sensitive to that shift. A 15 nm change in resolution moves every model, so recalibration is required on a different instrument.

8.6 No mineral index. We searched for a hydroxyapatite band, choosing candidates without yields from peaks shared across bones. The candidate apatite O-H band near 1430 nm tracks the 1930 nm water band (r = 0.89). Carbonate near 2345 nm shows a net absorption in only 8 % of bones. The strongest candidate, a sharp feature at 1384 nm, moves with a twin at 1883 nm (r = 0.84) and ignores liquid water, the signature of water vapour. It still tracks collagen within Zafarraya (r = -0.60), possibly through pore air in more porous, degraded bone. A mineral/collagen ratio is also unstable by construction: its collagen denominator vanishes at the low yields where screening matters.

8.7 The Lugli NDI across sites. Applied to the 176 bones, the NDI correlates with yield at r = -0.05, and its test RMSEP is 5.05 % against 1.93 % for CI2045. Absorbance at 1959 nm correlates with the water band at r = 0.94. The 2195 nm collagen band is sound: with a shoulder reference at 2225 nm instead of 1959 nm, the same two-band form reaches J = 1.84 %. The NDI may work within one bone or one uniform set, where water is roughly constant, but not as a calibrated index across collections.

8.8 Detecting and compensating PVA. PVA can be flagged reliably from the second derivative at 2137 nm, provided the dip is read against what the bone's collagen predicts there. At Zafarraya, bones carrying PVA differ from clean bones of the same yield by

second-derivative dips at 2138, 2253 and 2302 nm and an upward peak at 2273 nm. These lie within 3-6 nm of the reported PVA peaks; the upward peak is the gap between the 2250 and 2296 nm bands, not a third band.

The raw dip at 2137 nm is not enough. In clean bones the second derivative there sits on a lobe that rises with collagen (r = 0.91), so in a collagen-rich bone the PVA dip is offset. Clean bones, however, lie on a tight line between the 2045 nm band and the 2137 nm value, and PVA pushes a bone below that line. The rule is therefore a residual:

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s_{\mathrm{PVA}} = -\left[ D_{2137} - \left( a + b \, D_{2045} \right)
\right]
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Here D is the Savitzky-Golay second derivative of pseudo-absorbance (window 31, cubic), computed on the full spectrum. The coefficients a and b are fitted on clean calibration bones, and a bone is flagged when its score exceeds their 99th percentile. Both are specific to the instrument and must be refitted on a new one.

We tested the rule by adding synthetic PVA bands of known depth to the 40 clean test bones, with thresholds set on the 100 calibration bones.

Rule	Clean bones flagged	PVA 0.0025	PVA 0.005	PVA 0.005, bones with at least 8 % collagen	PVA 0.01	Zafarraya bones flagged
Raw dip at 2137 nm	0 %	0 %	40 %	0 %	70 %	12
Dip at 2137 nm against the collagen-expected value	**0 %**	**30 %**	**100 %**	**100 %**	**100 %**	**12**
Upward peak at 2273 nm	2 %	18 %	22 %	0 %	55 %	9

PVA depths are the absorbance of each synthetic band. At Zafarraya the corrected rule agrees with a two-band flag (dips at 2135 and 2250 nm) on 34 of 36 bones. Adding the 2273 nm peak to the rule made detection worse in rich bones, because collagen also pulls that value down.

PVA can also be compensated. Projecting a basis of synthetic PVA shapes out of the second-derivative spectra before PLS, a one-direction form of external parameter orthogonalisation, leaves the 2030-2060 nm model unchanged: PVA has no bands there. On the wider 2000-2300 nm range it cut the error on the 14 PVA-flagged Zafarraya bones from 1.92 to 1.21 % and halved their under-prediction (-1.24 to -0.58 %), at no cost on clean bones. That test rests on few, low-collagen bones.

One caution remains. At equal yield, PVA bones show a slightly shallower 2045 nm band (95 % bootstrap interval +0.2e-5 to +4.7e-5, about a third of the spread between bones). PVA does not absorb there, so the likely causes are optical dilution by the coating or collagen lost to solvent treatment. The 2030-2060 nm model is immune to PVA bands, not necessarily to a PVA coating.

9. Open questions and research gaps

The literature has established that NIR detects collagen. It has not yet established how well a calibration transfers to a new site, which is the question a prescreening campaign actually faces.

1. **Prospective validation.** Every quantitative model to date was evaluated on bones whose yields were already known, and usually extracted before scanning. No study has scanned an assemblage, frozen its predictions, and only then extracted collagen to test them. Nor has any compared NIR, Raman and ATR-FTIR on the same bones and spots, although each is weakest near the 1 % threshold and Madden et al. (2018) suggest that combined pass/fail rules may be more reliable than any single test.
2. **Site offsets.** Offsets of 1-2 percentage points appear at Zafarraya and among the Spanish reference bones, but their cause is unknown. Candidates include burial environment, porosity, mineral alteration, residual consolidants and differences in extraction. Local recalibration with a handful of bones reduces them, but how many bones are needed, and how to choose them, has not been tested.
3. **The low-yield range.** The 1 % threshold is where decisions are made, yet bones under 1 % were deliberately thinned out of the largest calibration set (Ryder et al., 2026). How well models separate 0.5 % from 1.5 % collagen is the least known part of their performance; Ryder et al. plan experiments on it.
4. **Representativeness of the scanned spot.** Heterogeneity within a bone is large (Section 7.2). The number and placement of scans that best represent the portion to be extracted is unknown for single-point instruments.
5. **The mineral phase.** No NIR band has been validated against an independent mineral measurement such as ATR-FTIR crystallinity or ash content. The assignment of the 1900-1960 nm region remains contested. Raman (phosphate at 960 cm-1) and ATR-FTIR (phosphate near 1015 cm-1) measure the mineral directly and could supply that reference if taken at the same spot as the NIR scan.

6. **Measurement conditions.** Room humidity, white-reference age and scan order are not reported, yet the re-analysis found vapour bands that follow collagen. Their role cannot be tested without these metadata.
7. **Transfer between instruments and protocols.** Calibration transfer between spectrometers and between extraction protocols (ultrafiltration, XAD resin, isotope-laboratory methods) is untested for bone.

10. Implications for the Spanish campaign

The campaign should be designed as the first prospective test of NIR prescreening: predictions frozen before extraction, scored against yields from the same spots, with the metadata needed to explain site offsets.

10.1 Questions and hypotheses

- **H1, transfer.** The published PLS on 2030-2060 nm and the CI2045 index, frozen before extraction, predict yield at new Spanish sites with an RMSE no worse than at Zafarraya (about 2.7 %), and classify bones at the 1 % threshold with sensitivity of at least 0.8 for the Sample decision.
- **H2, offset.** Spanish sites share a negative offset of about 1.4 % at equal band depth, as in the reference set. The alternative is that offsets are site-specific, driven by burial environment as in NW Spain (Martínez Cortizas and López-Costas, 2020).
- **H3, local correction.** Extracting five bones per site and subtracting their mean error reduces RMSE at that site by at least 0.5 percentage points.
- **H4, representativeness.** Averaging three spots per bone reduces error compared with one spot.
- **H5, combined screening.** A rule combining NIR with Raman and ATR-FTIR at the same spot classifies bones at the 1 % threshold better than any single method.

10.2 Sample size

With a residual SD near 1.6 %, estimating a site offset to within about ±0.3 % (one standard error) needs roughly (1.6 / 0.3)^2, or about 30 bones per site. The test set of 40 bones gave a 95 % interval for RMSE about 0.9 percentage points wide (1.44-2.36 % for CI2045), so totals well above 40 are needed to resolve differences between models. The yield range should be spread deliberately, with at least a third of bones expected near the 1 % threshold.

10.3 Measurement protocol

1. Record each bone's conservation history: any consolidant, adhesive or solvent treatment, with dates where known.
2. Clean the surface dry; photograph it with a scale; mark three flat, non-overlapping spots on the periosteal surface.
3. Condition bones in the measuring room for at least 24 h, and log temperature and relative humidity at each session.
4. Take a white reference at the start of each session and at fixed intervals. Log its time and the scan order of every bone.
5. Acquire 50 co-added scans per spot with the probe flush and perpendicular, as in Ryder et al. (2026), and keep the raw reflectance files.
6. Run quality checks before any prediction: the PVA rule of Section 8.8 (second-derivative dip at 2137 nm read against the 2045 nm band, with the upward peak near 2273 nm as a visual check); sharp vapour dips at about 1384 and 1883 nm; Hotelling T2 and Q residuals against the reference set.
7. Compute and archive predictions from the frozen models (PLS and CI2045), with a timestamp, before any extraction.
8. Extract collagen from the scanned spots by the same ultrafiltration protocol, in one laboratory, recording the >30 kDa and total fractions.
9. Before drilling, take Raman spectra on an adjacent fresh break or cross-section. Then measure whole-bone %N and ATR-FTIR on the drilled powder, and ATR-FTIR again on the extracted collagen (Section 10.6). This gives an independent mineral reference and a direct comparison of all methods on the same spot.

10.4 Metadata to record

Field	Why it matters
Site, context, stratigraphic unit, approximate age	Site offsets and diagenesis (Sections 7.6, 8.3)
Burial environment: soil pH, drainage, cave, open air or dune	The main preservation driver found in NW Spain (Martínez Cortizas and López-Costas, 2020)
Element, tissue (cortical or trabecular), spot coordinates	Heterogeneity and scan representativeness (Section 7.2)

Field	Why it matters
Conservation and solvent history	Consolidant interference and collagen loss (Sections 7.5, 8.8)
Room temperature and relative humidity, scan order, white-reference time	Water and vapour effects (Sections 8.6, 9)
Instrument serial, wavelength check, resolution, for each spectrometer	Calibration transfer (Sections 4, 8.5)
Extraction protocol, laboratory, >30 kDa and total yields, C:N of the extract	Definition of yield and quality (Sections 2, 7.3)
Whole-bone %N; ATR-FTIR and Raman spectra and indices	Benchmark against established methods; mineral reference (Sections 2, 9, 10.6)

Spectra should be archived in an open format (CSV or JCAMP) with every reference value, as Madden et al. (2018) recommend, so that results can be compared between laboratories.

10.5 Analysis plan

The primary analysis scores the frozen predictions against measured yields: RMSE, bias, and screening sensitivity and specificity at 1 % (truth) using the Skip / Check / Sample cuts of 1.75 and 2.95 %. Secondary analyses estimate the offset of each site with its interval, the gain from local correction as a function of the number of bones extracted, the effect of one against three spots, and the role of humidity and vapour bands. Only after this, a recalibration including the Spanish bones is fitted and evaluated by leave-one-site-out validation, so that the published model and the new one are compared on bones neither has seen. For H5, each method's classification at 1 % and their combinations are compared on the same bones, with the same leave-one-site-out scheme.

10.6 Raman and ATR-FTIR protocols

Raman and ATR-FTIR see the mineral phosphate directly, which NIR cannot, so they supply the mineral/collagen ratio that no NIR band could validate (Section 8.6). The protocols below follow the four studies that tested them on archaeological bone.

ATR-FTIR on bone powder (after Lebon et al., 2016)

1. Drill about 50 mg at the NIR spot, grind finely, dry all powders the same way (for example overnight in a desiccator), and press under 1 mg onto a diamond crystal.

2. Acquire 4000-400 cm-1 at 2-4 cm-1 resolution with 64 scans, on three separate aliquots per bone.
3. Adjust the anvil pressure until the phosphate band near 1015 cm-1 reaches an absorbance of about 0.5. Lebon et al. found this normalisation essential for reproducible ratios.
4. Compute the amide I / phosphate ratio from band areas: amide I 1710-1590 cm-1 with a linear baseline, and phosphate 1110-940 cm-1 with a baseline from 1160 to 890 cm-1. Their calibration, collagen (wt %) = 113.13 × ratio + 1.69, can serve as a first estimate, but it should be refitted against the campaign's own yields. Expect little information below about 3 % collagen.
5. Check for humic bands near 1715 and 1602 cm-1. Where FTIR and %N disagree, re-measure a subset after a NaOH wash, as Lebon et al. did: nitrogen-rich contaminants inflate %N but hardly affect FTIR.
6. Record standard mineral indices as well, such as crystallinity (splitting factor) and carbonate to phosphate. They are not covered by the studies reviewed here, but they provide the mineral-alteration measure that NIR lacks.

ATR-FTIR on extracted collagen (after Martínez Cortizas and López-Costas, 2020)

1. Acquire 4000-400 cm-1 at 4 cm-1, and locate bands from second-derivative minima.
2. Compute the carbohydrate / amide I index (985-1140 over 1590-1720 cm-1), the CH2 / amide II index (1338 over 1490-1590 cm-1) and the 1660 / 1690 cm-1 cross-link ratio, and compare them with C:N before dating or isotope work. In well-preserved collagen the first two indices rise together; a departure from that trend marks degraded structure.

Raman (after France et al., 2014; Madden et al., 2018)

1. Use 1064 nm excitation to limit fluorescence.
2. Measure a freshly exposed cross-section or a natural break, cleaned of sediment. Treat outer surfaces only as a first look, since they misclassify well-preserved bones.
3. On a benchtop FT-Raman: 1024 co-added scans at 4 cm-1, rising to 2048-4096 for dark or collagen-poor bone, with laser power kept below burning (France et al. used up to 0.13 W on a 50 um spot). Do not smooth the spectra; it rarely helped.
4. Take three spectra per spot and keep the best after a visual check for noise, fluorescence and extraneous peaks.
5. Measure peak heights and baselines as averages over short windows: phosphate 958.5-961.5 cm-1 (anchors 1135-1175 and 890-920), CH2 1448.5-1451.5 (anchors 1510-1550 and 1130-1170), amide I 1634.5-1637.5 (anchors 1730-1770 and 1530-1570).

6. Report 960/1636 and 960/1450 and read them as likelihood bands rather than single cut-offs. On handheld cross-section spectra, 960/1450 below 9.3 was always well preserved, 9.3-18.3 was 75 % well preserved, and above 21.5 never (Madden et al., 2018). On a benchtop, 960/1636 up to 19.4 marks well-preserved collagen (France et al., 2014). A negative ratio indicates a poor bone or contamination, and calls for a new measurement.
7. On bones flagged for PVA (Section 8.8), rely on 960/1636 only: the 1450 cm-1 band is shared with consolidants. Resolving 1636 cm-1 reliably needs a benchtop instrument.

11. Conclusions

NIR spectroscopy is ready to be tested as a routine, non-destructive collagen screen, but not yet to be trusted at a new site without local checks.

- **The signal is understood.** Collagen is read through protein combination bands at 2030-2060, 2150-2195 and 2270-2300 nm. A one-component PLS on 2030-2060 nm is as good as any alternative tested, and a three-wavelength index centred on 2045 nm comes within 0.15 percentage points of it on held-out bones.
- **Accuracy is limited by the reference, not the spectrometer.** Errors within a collection (1.2-2.2 %) approach the 1.5-1.6 % precision of extraction. Errors at a new site (about 2.5-2.8 %) are dominated by offsets whose causes are unknown.
- **The mineral cannot yet be read in NIR.** The band assigned to hydroxyapatite at 1959 nm behaves as water across many sites, and no other candidate passed physical and statistical checks. Mineral information should come from Raman or ATR-FTIR, which see phosphate directly; both screen collagen reliably only from about 3 %, so neither replaces NIR at the 1 % threshold, but together they can be tested as a combined screen.
- **Screening is the right use.** Decisions at the 1 % threshold, with an uncertain middle band and local correction, are better supported than precise yields.

The Spanish campaign can close the most important gap: a prospective, frozen-prediction test at new sites, with the metadata needed to explain why sites differ.

References

This review is based on ten primary studies read in full: six on NIR, two on Raman and two on ATR-FTIR. It is not a systematic search of the literature. Works marked as secondary were not consulted directly; their content is reported as cited by the primary studies.

Primary studies

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Data and code

- Spectra, yields and code of Ryder et al. (2026): github.com/gerlis22/NIR_Collagen and the article's supplementary data.
- Re-analysis (Section 8): notebooks `01_framing_and_data_audit` to `08_mineral_index` in the `model_exploration/` folder of the project repository, with results cached in `model_exploration/results/`.